AI Is Delivering Micro Advances in Healthcare

While Missing the Macro Opportunity

A national information architecture to reduce costs, improve treatment decisions, and measure outcomes while protecting patient privacy

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This document examines whether a national healthcare information architecture could create a resolved longitudinal history for each patient that enables AI to improve treatment decisions, reduce administrative effort, identify improper or fraudulent payments, and measure healthcare outcomes.

It does not recommend immediate development. It proposes a focused 60-day assessment by six experts to evaluate technical feasibility, measurable benefits, privacy and security requirements, legal authority, political viability, public trust, and the relationship to TEFCA and existing healthcare information systems. Existing studies identify potential annual savings measured in tens of billions of dollars from administrative simplification and payment integrity alone—before considering the value of improved treatment decisions and healthcare outcomes.

At a Glance

The problem: Healthcare information is dispersed among providers, payers, pharmacies, laboratories, imaging centers, and incompatible record systems. Patients repeatedly reconstruct their histories while clinicians make decisions without convenient access to all relevant information.

The opportunity: Create a secure national information architecture that maintains a resolved longitudinal history for each participating individual. Existing healthcare systems would continue to collect information and support providers while exchanging validated information through the national architecture.

The role of AI: AI would review the volume of clinical notes, tests, images, treatments, and outcomes to identify information relevant to the current decision. It would support—not replace—the judgment of healthcare professionals.

The potential value: Existing studies indicate that annual savings in the tens of billions of dollars can be achieved by reducing administrative effort, incomplete authorizations, payment errors, and fraud. Additional value could come from fewer repeated tests, earlier diagnoses, better treatment decisions, and improved healthcare outcomes.

The safeguard: Individuals would control authorized access to their information and see who accessed it and when. Access would be limited by purpose, fully audited, and protected by federal penalties for unauthorized use.

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Background:

Micro Advances, Macro Opportunity

We regularly hear about narrowly targeted uses of artificial intelligence in healthcare.

AI models can help identify suspicious areas in medical images, compare current scans with earlier ones, and draw a physician's attention to changes that might otherwise be missed. AI can help clinicians prepare more complete notes while spending less time typing and more time with patients.

These can be important—even life-changing—advances. But they apply AI to individual tasks within the healthcare system. They do little to address fragmented information and administrative friction that affect the entire system.

In fact, a micro improvement for one participant does not necessarily reduce the total cost of healthcare. More complete clinical documentation may justify more accurate—and sometimes higher—reimbursement. A provider operating under negotiated payment rates may benefit from greater productivity, but those savings may not reach the insurer or the patient. Claims of savings may also not include the full costs of development, integration, infrastructure, maintenance, and oversight.

A recent systematic review found promising economic outcomes from clinical AI applications, including fewer unnecessary procedures and more efficient use of resources. But it examined only 19 studies across selected applications and called for more standardized, long-term economic evaluation. That is evidence that some AI applications can be cost-effective—not yet evidence that adding AI throughout today's healthcare system will reduce its total cost. Systematic review

Meanwhile, a much larger opportunity to reduce cost and improve effectiveness is sitting in front of us.

Sometimes literally.

Part of the Cost Was Sitting in My Lap

I will turn 76 this year. Since moving to Florida, I have had four primary-care physicians. This year, my newest one referred me to a cardiologist and an orthopedic specialist.

At each office, I was handed a clipboard and asked for nearly the same information.

Once again, I tried to remember the dates of surgeries and injuries. Once again, I worked through pages of medical conditions, checking yes or no. Much of the information already existed in the records of my previous physicians—but apparently "somewhere in my medical history" was not the same as "available here."

While I completed one set of forms, the waiting-room television carried another news story about the rising cost of healthcare.

I looked at the pages on my clipboard and wondered who would enter this information, whether anyone would reconcile it with my other records, and what would happen if my 76-year-old memory supplied the wrong answer.

CRITICAL INFORMATION		
Adhesive tape / latex allergy	▫ Yes	▫ No
Anticoagulant treatment	▫ Yes	▫ No
Artificial heart valves	▫ Yes	▫ No
Artificial joint	▫ Yes	▫ No
Bacitracin / Neosporin allergy	▫ Yes	▫ No
Bleeding disorders	▫ Yes	▫ No
Breast cancer / Other cancer	▫ Yes	▫ No
Epilepsy	▫ Yes	▫ No
Epinephrine sensitivity	▫ Yes	▫ No
…		

Then I looked beyond the receptionist at the people working on scheduling, insurance verification, claims, prior authorizations, payments, and medical records for a practice with two physicians.

As the television continued asking why healthcare costs so much, I began to suspect that part of the answer was sitting in my lap.

While AI contributes to micro improvements in healthcare, it could also attack enormous macro opportunities:

- reduce provider–payer administrative effort;
- prevent unnecessary duplication of tests and treatments;
- use complete longitudinal information to help clinicians make more accurate decisions and keep people healthier.

We are using AI to improve individual healthcare tasks while overlooking the information architecture needed to improve healthcare as a system.

Why I Am Raising the Question

I am not a doctor, and I don't play one on TV. I do claim considerable expertise in determining whether an information architecture can deliver the information required to support better decisions and outcomes.

Over my career, I participated in four-to-eight-week benchmark engagements with more than 100 major organizations across many industries, including several involved in healthcare. The technology varied, but the fundamental question did not:

> **Could the organization get the right information to the right person in time to make a better decision?**

In healthcare, that question can affect both cost and survival.

When Existing Information Is Not Available

The two personal examples establish two different failures:

- The accumulated history exists but is too fragmented to guide continuing treatment.
- A specific technical fact exists but is unavailable at the moment of treatment.

A friend of mine has been battling disabling intestinal problems for nearly four years. Different specialists have ordered scans and colonoscopies, prescribed medications, and recommended specialized physical therapy. Yet the periodic episodes continue.

Were the previous tests, treatments, and results available to the current specialist in a form that could be readily understood? Had the healthcare system learned from each unsuccessful intervention—or merely recorded it somewhere? Could the experience of specialists treating comparable patients offer another course of action?

When my stepson was receiving chemotherapy through a port in his chest, the staff preparing an injection asked him what size needle had been used previously.

Why was the patient being asked to supply a technical detail needed to treat him safely? What could happen if he gave the wrong answer?

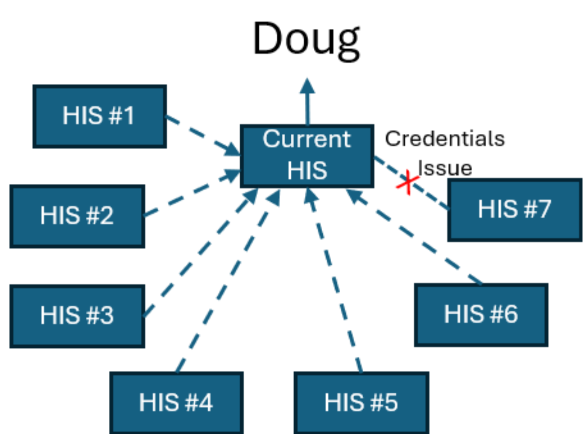
The information existed. It simply was not available when it was needed.

More Data, Less Usable History

Our advanced technology is generating more health data than ever. At the same time, increasing specialization is dispersing that information among more providers and more systems. Each specialist may successfully treat the part of the patient assigned to them while no one has a complete view of the patient.

When I began seeing my new primary-care provider this year, I updated my information in the practice's **H**ealth **I**nformation **S**ystem (HIS) and linked six other healthcare information systems to my account. I could not link a seventh due to issues with my login credentials.

That does not include records from three dental providers and two dental specialists I have used during my adult life.



Connected - But not integrated

Technically, I have created pathways to much of my history. That does not mean it has been integrated, reconciled, or made useful.

For a new provider to understand my history, someone would have to search years of physician notes, test results, diagnoses, medications, and procedures distributed across those systems. Identifying the information relevant to a new condition could require an hour or more of searching and study—time most providers do not have.

This is not a modern information architecture designed for people who change jobs, move to new cities, and receive care from multiple primary-care providers and specialists.

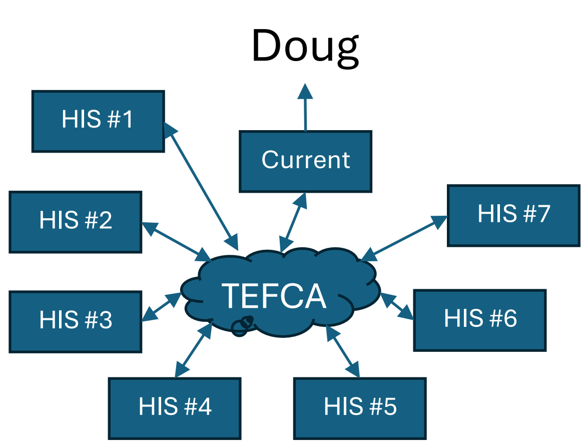
It does not create a reliable longitudinal history in which identities have been matched, duplicate records recognized, conflicting information identified, and important facts validated.

Without that foundation, AI cannot reliably extract the history relevant to a current decision, identify potentially redundant tests and treatments, compare results over time, improve authorization decisions, or evaluate costs and risk-adjusted outcomes across providers.

AI does not replace the judgment of healthcare providers. It can perform the work that no provider has time to perform: continuously examining years of distributed information and presenting the relevant evidence when decisions are made about diagnoses, treatments, referrals, and prescriptions.

TEFCA: Important Progress, Not the Complete Foundation

The federal government recognizes part of this problem. The Trusted Exchange Framework and Common Agreement—TEFCA — establishes a nationwide framework through which participating healthcare organizations can query and exchange patient information. As participation expands, TEFCA may eliminate some of the portal linking, passwords, and manual record transfers patients experience today.



Exchanged - but not reconciled

That is important progress, but exchanging records is not the same as integrating information.

TEFCA does not, by itself, create a continuously maintained longitudinal history in which records from different sources have been matched, duplicate events identified, conflicting information reconciled, and outcomes connected to treatments. It does not create the analytical foundation needed to compare costs and risk-adjusted outcomes across providers. Nor does it prepare the relevant facts for a physician before a treatment decision is made.

TEFCA could provide an important channel for information to reach such a foundation. It is not the complete foundation.

The macro opportunity requires an information architecture organized around the patient and the improvement of healthcare—not around the institutions that currently possess fragments of the patient's history.

Recommendations:

1. Create a trusted longitudinal health history organized around each patient.
2. Use national standards to collect clinical, claims, laboratory, imaging, pharmacy, and other relevant information.
3. Preserve the original source of every fact while identifying duplicates, conflicts, and missing information.
4. Use AI to present relevant history and evidence to providers at the point of decision.
5. Automate routine information exchange between providers and payers.
6. Use risk-adjusted outcomes to compare treatments and providers serving comparable patients.
7. Give patients control, visibility, and a complete audit trail of access to their information.
8. Permit approved analysis of protected data without exposing individual identities unnecessarily.
9. Use TEFCA and existing healthcare systems as information sources rather than expecting them to perform the integration.
10. Measure success by reductions in administrative effort, avoidable care and adverse outcomes—not by the volume of data exchanged.

Discussion:

Benefits to the Individual:

The individual—not the provider, payer or government—should be the principal beneficiary of a patient-centered information architecture.

Today, patients repeatedly supply the same information, search for records, remember prior treatments and carry technical details from one provider to another. Yet they generally cannot see a complete history, determine who has examined it, or easily compare the results achieved by different providers.

A Usable, Patient-Controlled History

A longitudinal health information architecture would give individuals:

- **One understandable health history.** Patients could see diagnoses, medications, procedures, test results, implanted devices, and significant events collected across providers.
- **Visibility into access.** Patients could see who accessed their information, when it was accessed, what information was retrieved, and the stated purpose.
- **Fewer unnecessary forms.** Patients would not have to reconstruct the same history from memory whenever they changed providers or are referred to specialists.
- **Fewer avoidable tests and scans.** Providers could see prior tests, retrieve the results, and determine whether repetition was clinically necessary.
- **Safer, better-informed treatment.** AI could identify relevant allergies, prior complications, unsuccessful treatments, medication conflicts, and changes in test results for provider review.
- **Faster, better-informed prior-authorization decisions**. Relevant clinical evidence could automatically accompany requests, reducing denials and resubmissions due to incomplete information.
- **Continuity when moving or changing providers.** A patient's usable history would not remain trapped in the city, health system or employer they left behind.
- **Better information for selecting providers.** Patients could compare costs, experiences, and risk-adjusted outcomes across providers treating people with comparable conditions.
- **A clearer understanding of treatment choices.** Cost information could be considered alongside outcomes, complications, recovery times and the experience of similar patients.
- **The ability to identify errors and conflicts.** Patients could flag incorrect information and see how providers resolved disputed or inconsistent entries.

Making Patient Choice Meaningful

The least expensive treatment is not cost-effective if it fails, causes complications, or requires repeat treatment. Patients need to compare the total cost of care with the outcomes achieved.

Patients are routinely told to become informed consumers of healthcare, but they are denied much of the information required to do so. A patient cannot intelligently compare providers when prices are opaque, outcomes are not risk-adjusted, and their own history is scattered across multiple systems.

Patient choice becomes meaningful only when the patient has usable information.

Better-Informed Prior Authorization

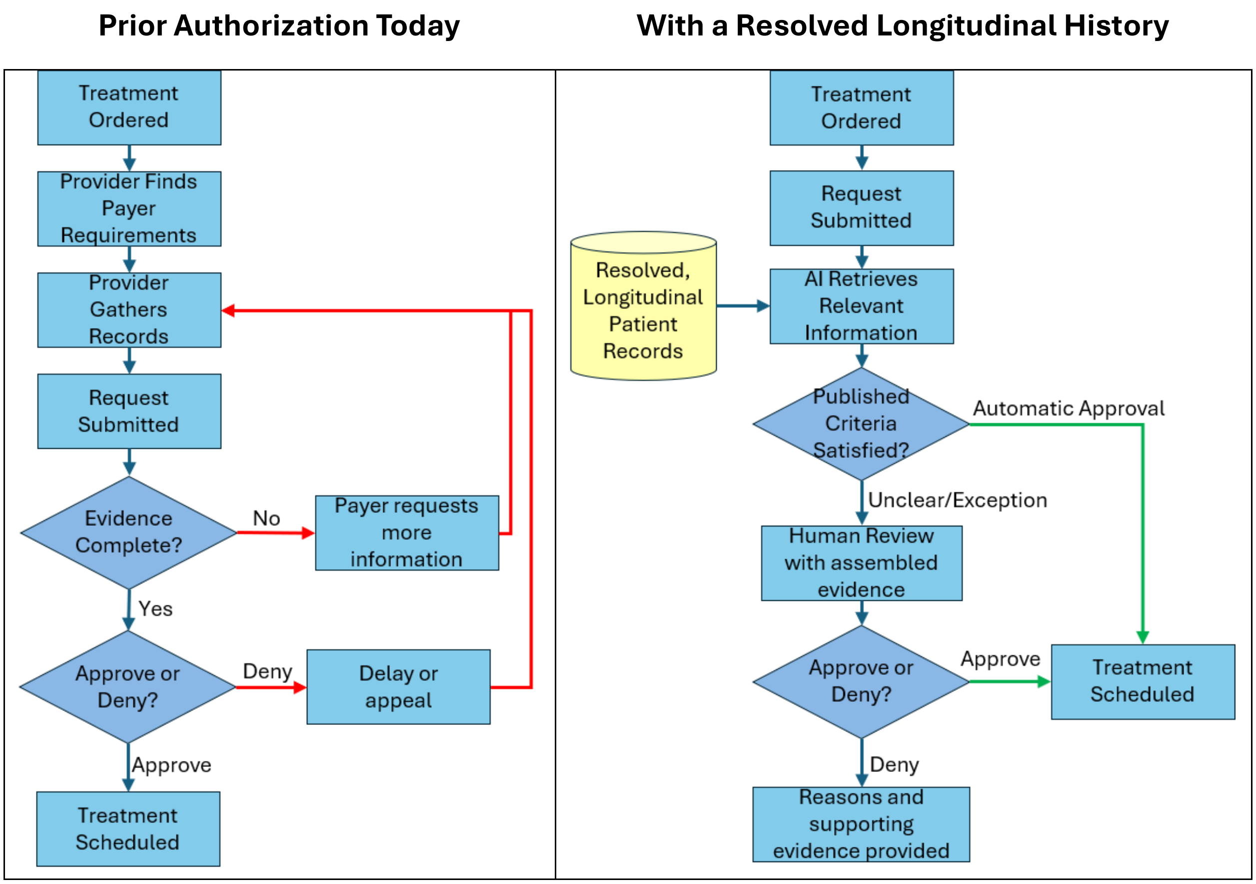
A significant pain point for the individual is the denial of prior authorizations and resulting delay of service.

Prior authorization is intended to prevent unnecessary or inappropriate treatment. In practice, a decision may be made without a complete understanding of the patient's history.

A patient may be denied an MRI and required to complete weeks or months of physical therapy first. That may be appropriate for some patients. It may be inappropriate for a patient whose history, symptoms, previous treatment, or risk factors indicate the need for imaging.

Today, the provider may need to manually assemble and resubmit the evidence. The patient waits while the provider and payer exchange forms, notes, and telephone calls.

A longitudinal information architecture could automatically provide the payer with the information necessary for the decision: prior physical therapy, medications, earlier imaging, relevant diagnoses, symptom progression, and the results of similar treatment decisions involving comparable patients.



Better information should produce both more appropriate approvals and more defensible denials. The objective is not to approve every request. It is to reduce decisions made from incomplete information.

Benefits to the Provider:

Healthcare providers do not suffer from a shortage of patient data; they suffer from insufficient time to find, reconcile, and review it. Relevant information may be buried in years of physician notes, test results, images, prescriptions, and records maintained by other organizations.

A resolved longitudinal history would allow AI to identify pertinent conditions, prior treatments, medication changes, and trends before a clinical decision is made. The same architecture could reduce administrative work related to authorizations, claims, referrals, and record exchange, allowing clinicians and their office staff to spend more time providing care.

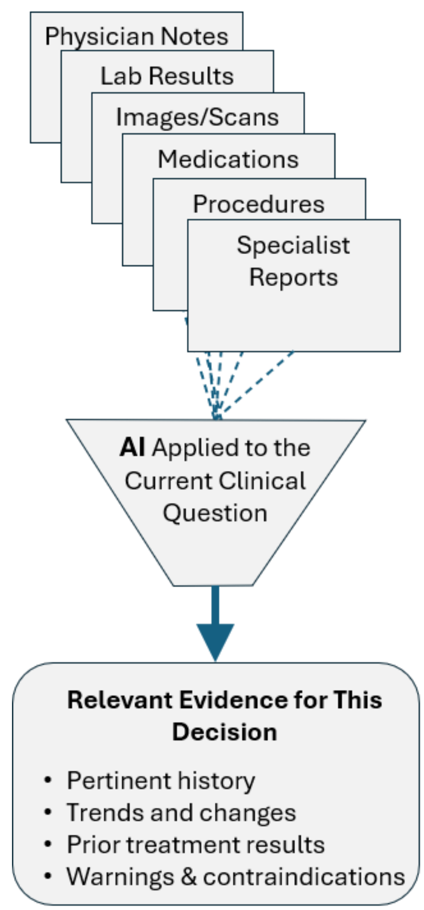
Relevant Information at the Point of Care

- relevant patient history presented before the encounter;
- trends and meaningful changes highlighted automatically;
- fewer hours spent searching external records and reading repetitive notes;
- identification of prior unsuccessful treatments, complications and contraindications;
- automatic comparison of current and previous tests and images;
- source links allowing providers to verify every AI-generated summary;
- more complete referrals and transitions between providers;
- automatically prepared prior-authorization documentation;
- fewer claim rejections caused by incomplete or inconsistent information;
- less manual handling of records, forms, coding and payer inquiries;
- more time for patients and clinical judgment.

Making the Accumulated Record Usable

Healthcare providers face a paradox: they have access to more patient data than ever, yet less practical ability to understand it all before making a decision.

An older patient with several chronic conditions may have decades of physician notes, laboratory results, scans, procedures, medications, and specialist reports. When those records are distributed across multiple systems, a provider may not even know what exists. When records are available, the volume can make comprehensive review impossible within the time allowed for an appointment.



In practice, the information most consistently visible may be the current medication list, allergies, active diagnoses, and recent results. Important background can remain buried in years of repetitive notes and disconnected test results.

A study of outpatient notes found that the notes increased by 60% in length between 2009 and 2018. By 2018, approximately 71% of the text was templated or copied rather than directly entered. Longer records therefore do not necessarily provide proportionately more useful information. They can make the important facts harder to find. JAMA Network Open

Reviews of electronic-record information overload have identified potential consequences including missed information, delayed care, and patient-safety risks. [AHRQ Patient Safety Network]

AI documentation tools may help clinicians produce notes more efficiently. They may also create more detailed notes, adding still more information for the next provider to review.

AI should not only help create the record. It should help make the accumulated record usable.

Reducing Medical-Office Administration

The same longitudinal history could reduce work throughout the medical office. Demographic and historical information would not have to be collected repeatedly. Referrals could automatically include the relevant history. Prior-authorization requests could be assembled with the diagnoses, earlier treatments, and test results required by the payer. Claims could be checked for completeness and consistency before submission. Denials could be matched with the evidence necessary for review or appeal.

This would not eliminate every administrative position. It would eliminate much of the searching, rekeying, reconciling, faxing, telephoning, and resubmitting that consumes staff time without directly improving patient care.

Providers do not need more data presented to them. They need the relevant evidence extracted from the data they already have—and from the data they currently cannot see.

Benefits to the Payer:

A longitudinal health information architecture would provide payers with timely, complete evidence for authorization and payment decisions, reduce administrative effort and improper or fraudulent claims, and help determine whether coverage policies reduce total costs while improving patient outcomes.

Better Information for Decisions

A longitudinal health information architecture could provide:

- More complete and accurate claims.
- Less manual review and reconciliation.
- Fewer requests for missing documentation.
- Faster, more consistent prior-authorization decisions.
- Fewer avoidable appeals and overturned denials.

- Better coordination when members change insurers.
- Improved detection of duplicate, inconsistent, or potentially fraudulent claims.
- Better identification of treatments producing the best outcomes for comparable patients.
- Risk-adjusted comparisons of provider cost and performance.
- Earlier identification of patients who may benefit from preventive or coordinated care.
- Evidence showing whether coverage policies reduce total cost or merely postpone it.

P~~P~~ayers possess broad claims histories, but claims primarily describe what was billed and paid. They frequently lack the clinical detail needed to understand why a service was requested, what had already been attempted, and what happened to the patient afterward.

Prior Authorizations • Claims • Payments

Limits of the Payer's Information

OFTEN HIDDEN OR DISCONNECTED

Clinical Context • Prior Treatment Results
Patient Outcomes • Costs Under Other Payers
Later Complications• Total Cost Over Time

That missing information creates work on both sides of the provider–payer interface. Payers request documentation, provider offices locate and submit it, and payers review it. Missing information leads to another request, a denial, or an appeal. The same evidence may be assembled and reviewed repeatedly by different people in different organizations.

A resolved longitudinal history could supply the relevant evidence electronically and in a consistent form. Claims could be checked against the documented diagnosis, treatment, and authorization before submission. Missing or conflicting information could be identified immediately rather than after the claim is rejected.

Prior authorization should become an evidence-based decision rather than a document-collection contest.

AI could assemble the symptoms, diagnoses, prior treatments, results, adverse effects, and applicable coverage rules. Straightforward requests could be processed consistently, while unusual, conflicting, or high-risk cases would be directed to qualified reviewers with the relevant evidence already assembled.

The standard should not be "AI decides." AI should assemble the evidence and apply explicit rules. Accountable people should remain responsible for exceptional or disputed decisions.

Measuring Total Cost and Outcomes

A payer can calculate how much was saved by denying an MRI today. It may have much less visibility into whether the patient later required emergency treatment, surgery, hospitalization, or months of ineffective therapy.

A longitudinal history could connect an authorization decision with the eventual cost and outcome—even if the patient changed providers or insurers. Payers could then distinguish between care that was unnecessary, care that was appropriately delayed, and care whose delay increased total cost or contributed to a worse outcome. They could also identify treatments that cost more initially but reduced subsequent complications.

This addresses a structural problem created when people change employers and insurance plans. One payer may bear the cost of preventive treatment while another receives the future savings. Conversely, delaying treatment may reduce one payer’s immediate expense while leaving a future payer, Medicare, or the patient with a higher cost.

Healthcare should not declare a decision cost-effective merely because its consequences occurred after the patient changed insurance companies.

Payment Integrity and Accountability

Claims and clinical evidence could be compared to identify duplicate billing, services unsupported by the clinical record, incompatible procedures, and unusual patterns across providers and payers. Such findings should trigger review rather than automatic accusations. An apparent inconsistency could result from fraud, coding error, incomplete information, or legitimate clinical circumstances.

Payers would receive better information and lower administrative costs, but they would also become more accountable. Approval times, denial rates, appeal reversals, repeated requests for information, and outcomes following delayed or denied treatment could be compared across payers.

Payers should welcome better information—but better information must create accountability on both sides of the transaction.

Evolution of Current Healthcare Information Systems

The United States has invested enormous amounts of money and effort in healthcare information systems. Replacing them would be financially unrealistic, operationally disruptive, and unnecessary.

These systems perform many functions, which are best managed close to the organization delivering care. Physician practices need systems that support appointments, clinical documentation, prescriptions, referrals, orders, billing, and patient communication. Hospitals need systems that support admissions, beds, laboratories, imaging, pharmacies, and thousands of specialized workflows. Payers need systems that support enrollment, benefits, claims, authorization, and payment.

Those responsibilities would remain.

What would change is the expectation that each system independently maintains the most complete available history of the patient.

A clearer division of responsibility

Existing healthcare systems	National architecture
Collect information during care	Assemble the longitudinal history
Support local clinical workflows	Match information to the correct patient
Schedule appointments and resources	Reconcile duplicate and conflicting information
Enter orders and prescriptions	Preserve provenance across sources
Capture physician and nursing notes	Identify clinically relevant history
Operate laboratories and imaging	Analyze trends across organizations and time
Submit and process claims	Compare risk-adjusted costs and outcomes
Communicate with local patients	Maintain national access and audit history
Continue operating during outages	Support population-level learning

The current systems would remain authoritative for the events they originate. A laboratory remains responsible for the result it produced. A physician practice remains responsible for its notes and diagnoses. A pharmacy remains responsible for the prescription it dispensed.

The national architecture would preserve those original assertions and create a resolved view across them.

Information would be submitted as care occurs

New diagnoses, medications, procedures, test results, and significant observations would be transmitted to the national architecture as they were created or updated.

The architecture would not wait until another provider requested a patient's records and then contact every prior system in real time. It would maintain the longitudinal history continuously.

If a source system were temporarily unavailable, its previously submitted information would remain available. When service is restored, new or corrected information would be synchronized.

Relevant longitudinal information should appear within the healthcare system the provider already uses.

Information Within the Provider's Existing Workflow

A cardiologist might receive a cardiovascular summary containing prior diagnoses, medications, imaging, procedures, and trends in relevant test results. An orthopedic specialist might receive a different view of the same longitudinal history. Both could examine the complete record when necessary and follow links to the original evidence.

The national architecture would determine what information might be relevant to the context. The existing system would present it as part of the provider's normal workflow.

This avoids forcing clinicians to learn another portal, remember another password, or search another database.

AI services: national and commercial

Existing healthcare systems could use AI services provided by the national architecture or integrate certified commercial models.

Competition should continue in the development of provider interfaces, workflow automation, clinical decision support, and specialized AI. Vendors would compete on how effectively they help providers use information—not on their ability to keep patient information confined within a proprietary system.

When a source corrects information, both the original and corrected versions would remain traceable. When two providers disagree, the national history would preserve both assertions and flag the conflict for resolution.

AI could identify inconsistencies, but only an authorized person or an established clinical rule could resolve them. The resolved history would always link back to the original evidence.

Resilience During Outages

Patient care must not be interrupted when the national architecture is temporarily unavailable.

Existing systems should retain a current, locally accessible summary for established patients, including medications, allergies, major diagnoses, implanted devices, and other information required for safe treatment. Emergency procedures would permit controlled access when identity or normal authorization could not be completed.

The national architecture would improve completeness without becoming a single point of operational failure.

What may diminish over time

As adoption increases, several existing activities should become smaller or disappear:

- repeated patient-history questionnaires;
- manual record requests and faxing;
- proprietary point-to-point interfaces;
- scanning outside documents into local records;
- manual reconciliation of medication and condition lists;
- repeated entry of demographic and insurance information;
- separate retrieval of records from multiple patient portals;
- provider staff assembling routine authorization packets;
- payer staff rekeying and validating submitted information.

The transition for vendors

Healthcare information-system vendors have significant investments in their products, customers, and specialized knowledge. They should be participants in the transition rather than treated as obsolete.

Vendors would adapt their systems to submit standardized events, receive resolved information, invoke approved AI services, and present relevant evidence to users. They could continue differentiating through workflow, usability, automation, specialty functionality, and service.

The change is not that current systems cease to matter. It is that no single provider's system should be expected—or relied upon—to represent the patient's complete history.

The national architecture would not replace the systems that operate healthcare organizations. It would provide the shared information foundation those systems have never been able to create independently.

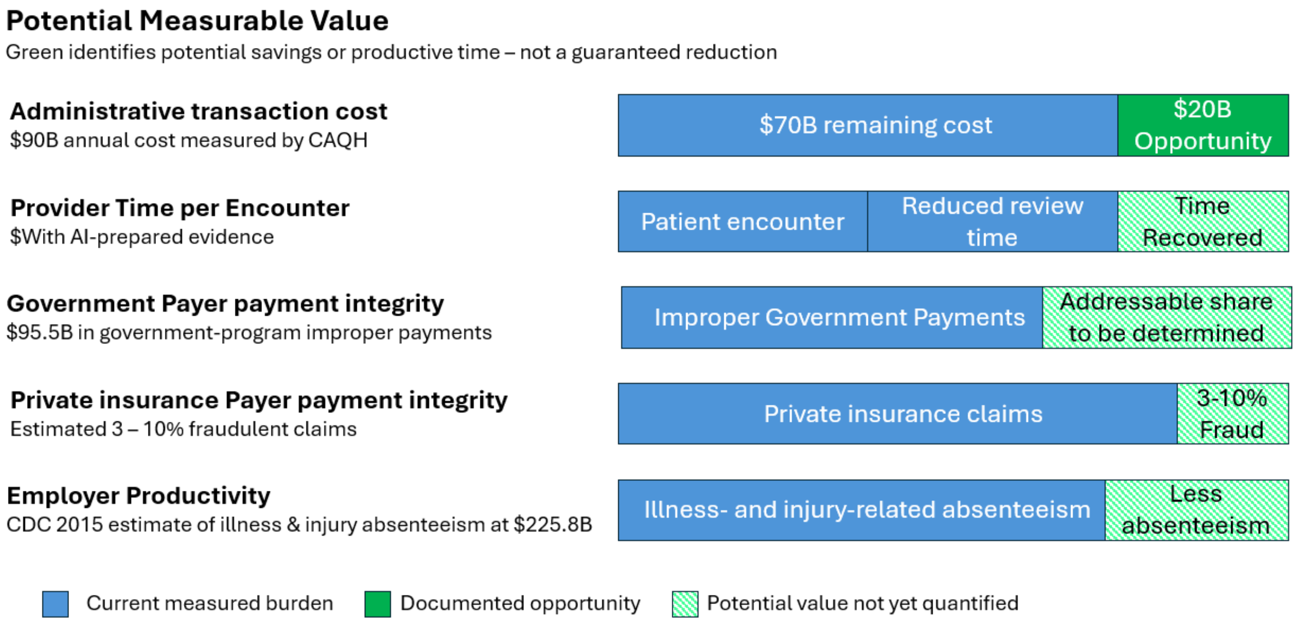
Reducing Administrative Waste and Fraud

The financial opportunity begins with the administrative exchange between providers and payers. The 2024 CAQH Index estimated that routine transactions such as eligibility verification, prior authorization, claims processing, and payment reconciliation cost approximately $90 billion annually, with about $20 billion potentially avoidable through greater automation. These estimates address primarily how information is exchanged, not the additional effort required to find, interpret, validate, and resubmit information scattered across multiple systems.

Improper payments suggest another substantial opportunity, although they should not be confused with fraud. For fiscal year 2025, CMS estimated approximately $95.5 billion in improper payments across Medicare fee-for-service, Medicare Advantage, Medicare Part D, Medicaid, and CHIP. These include documentation deficiencies, coding and eligibility errors, overpayments, underpayments, and payments that may involve fraud. GAO cautions that improper-payment estimates are not designed to measure fraud and cannot be used to estimate its extent.

The National Health Care Anti-Fraud Association cites estimates ranging from 3% to 10% of total healthcare spending, but no validated national estimate of fraud across government programs and private insurance exists. A resolved longitudinal health history would allow authorized systems to connect what was ordered, what was authorized, what service was documented, what result was produced, what was billed, and what was paid. That could reduce incomplete authorization requests, unnecessary denials, corrected claims, appeals, duplicate payments, billing unsupported by clinical records, and suspicious activity spread across multiple providers and payers.

Better healthcare can produce value outside the healthcare system. Earlier diagnosis, more effective treatment, fewer complications, and faster recovery could reduce employee sick days, disability leave, and impaired productivity while working. At the March 2026 average private-industry compensation cost, each avoided eight-hour absence represents approximately $373 of productive capacity. The amount attributable to the proposed architecture cannot yet be estimated, but employer productivity and disability outcomes should be included when evaluating its potential benefits.



These estimates cannot simply be added together because the categories overlap, and not every improper payment represents a recoverable loss. They do, however, establish a potential savings range large enough to warrant serious evaluation.

Privacy Implications

A national longitudinal health history would become one of the most valuable—and most dangerous—collections of personal information ever created. Privacy and security cannot be added as features after the architecture is built. **<u>They must determine how it is built, who can use it, and what happens when that trust is violated.</u>**

Distributed Does Not Mean Private

Today, a person's health information is distributed among physician practices, hospitals, laboratories, pharmacies, imaging centers, insurers, clearinghouses, cloud providers and government programs.

That distribution may limit the amount exposed by some individual breaches. It also creates more copies, more interfaces, more credentials and more people capable of accessing portions of the information. Each organization may apply different security practices and maintain different audit records.

Most patients cannot determine who examined their information, what was viewed, why it was accessed, or whether it was subsequently copied elsewhere.

Fragmentation is not the same as privacy.

A consolidated architecture increases concentration risk, so the design should be logically integrated yet **physically segmented**. No single stolen credential, system administrator, or compromised server should expose an entire national database.

Access Based on Purpose

Access should not be granted simply because someone works for a provider, payer, or government agency. It should be limited by role, relationship, purpose and time.

User	Permitted access
Individual	Complete history, corrections, disclosures and access log
Treating provider	Information reasonably relevant to treatment
Specialist	Relevant history plus requested supporting detail
Emergency provider	Necessary emergency information through audited "break-glass" access
Medical-office staff	Minimum information needed for scheduling, referrals, authorization or claims
Payer	Minimum clinical and administrative information necessary for coverage, authorization and payment
Quality analysts	Deidentified or strongly protected information
Researchers	Approved data appropriate to the study, normally deidentified
System administrators	Technical access without routine ability to view identifiable clinical content
Government personnel	No access merely because the government regulates or funds the architecture

Every authorization should expire when the treatment, employment, or administrative relationship ends.

Every Access Should Be Visible

HIPAA already requires covered organizations to employ audit controls, but it does not provide patients with a simple, comprehensive history of every routine access for treatment, payment, and health care operations. The proposed architecture should go further.

Every retrieval, display, download, transmission, correction, and AI analysis involving identifiable information should create a tamper-resistant record containing:

- ✓ who accessed the information;
- ✓ the organization represented;
- ✓ the date and time;
- ✓ the information accessed;
- ✓ the declared purpose;
- ✓ the authority permitting access;
- ✓ whether information was downloaded or transmitted;
- ✓ whether AI processed the information;
- ✓ whether emergency access was invoked.

Individuals should be able to review this history as easily as they review credit card transactions. Unusual access should generate an alert.

If a nurse in Florida accesses my record while treating me, I should be able to see it. If someone in another state accesses it at two o'clock in the morning with no treatment, payment, or operational relationship to me, both the system and I should want an explanation.

Permitted and Prohibited Uses

Identifiable information should be used only for the individual's treatment, payment, and authorized healthcare operations. Broader analysis should use deidentified or strongly protected data to improve healthcare quality, public health, and approved research.

Information collected for healthcare must not be repurposed to disadvantage the individual.

Federal legislation should specifically prohibit use of the architecture for:

- ⊗ employment decisions;
- ⊗ credit decisions;
- ⊗ marketing without explicit authorization;
- ⊗ sale of identifiable information;
- ⊗ unrelated commercial profiling;
- ⊗ life or disability insurance underwriting without explicit authorization;
- ⊗ political targeting;
- ⊗ unauthorized surveillance;
- ⊗ unrestricted law-enforcement searches;
- ⊗ discrimination based on predicted future illness;

- ⊗ training commercial AI models outside approved purposes;
- ⊗ reidentifying protected analytical data.

Any legally compelled access should require narrowly defined authority, retrieve only the necessary information, and leave a permanent audit record. The document can recommend stricter judicial oversight without attempting to resolve every legal exception.

Congress should enact legislation specifically governing the national health information architecture. Unauthorized access, intentional misuse, sale, alteration, destruction, or reidentification of protected information should constitute a federal crime, whether committed by an outside hacker, healthcare employee, payer, contractor, or government official.

The legislation should include:

- § criminal penalties proportionate to intent and harm;
- § greater penalties for trusted insiders and organized attacks;
- § personal liability for knowingly misusing access;
- § mandatory prosecution referral standards;
- § civil penalties for negligent security;
- § compensation and remedies for affected individuals;
- § whistleblower protection for reporting misuse;
- § mandatory breach notification;
- § independent investigation and public reporting;
- § prohibition against contractual waivers of patient rights;
- § uniform national minimum protections while allowing stronger state protections;
- § penalties for organizations that conceal breaches or fail to review suspicious access.

Security Architecture

The technical safeguards should include:

- ♦ zero-trust authentication;
- ♦ multifactor authentication resistant to phishing;
- ♦ encryption in transit and at rest;
- ♦ separate encryption keys for segmented data;
- ♦ isolation of identifiers from clinical content where practical;
- ♦ purpose-based and time-limited access tokens;
- ♦ continuous detection of unusual access;
- ♦ independent penetration testing;

- immutable, separately protected audit records;
- rapid credential revocation;
- geographically separated recovery systems;
- local availability of essential clinical summaries during outages;
- no single administrator capable of accessing or exporting everything;
- multiple-person approval for extraordinary data operations.

AI systems should receive only the data required for an approved task. Prompts, outputs, and model access should be logged. A model should not retain identifiable information for unrelated training.

The Privacy Trade

No architecture can promise that health information will never be attacked or misused. Today's distributed systems cannot make that promise either.

The appropriate question is whether a national architecture can provide stronger security, narrower access, more consistent enforcement, and far greater visibility than the fragmented environment that exists today.

Privacy should not mean that my information is scattered in a way that neither my physician nor I can find it. Privacy should mean that the right information is available for my care, unavailable for improper purposes and accompanied by a permanent record of everyone who used it.

A national health information architecture would require the public to place substantial trust in it. That trust must be earned through technical design, enforceable restrictions, individual visibility and criminal accountability—not requested through another eight-page form.

Next Steps

This document does not recommend immediately funding, designing, or building a national healthcare information architecture. It recommends a focused 60-day assessment by six people with the experience necessary to determine whether the concept is technically feasible, economically significant, legally supportable, politically viable, and worthy of further consideration.

The assessment should not become a large interagency committee. A small team should solicit evidence from the organizations that operate healthcare, exchange its information, regulate it, and experience its failures.

The six-person team should collectively provide expertise in:

- national-scale information architecture and data resolution;
- *healthcare delivery and clinical workflow;*
- *payer operations, prior authorization, claims, and payment integrity;*
- *EHRs, interoperability, TEFCA, and healthcare-information standards;*
- *privacy, cybersecurity, civil liberties, and healthcare law;*
- *bipartisan federal health policy, legislative implementation, and public trust*.

Patients, employers, state officials, researchers, and other affected interests would be represented through the interview process rather than by expanding the core team.

Structured Executive Interviews

The assessment should use structured, recorded interviews with knowledgeable representatives of healthcare providers, payers, EHR vendors, QHINs, federal and state agencies, patient and privacy organizations, employers, researchers, and congressional or executive-branch policy experts. Interviews could be conducted remotely to reduce time and travel. With permission, they should be recorded and transcribed.

The initial questions should follow the business-discovery structure:

1. What are the three to five most important problems or opportunities?
2. What information is needed to address each one?
3. What could be done differently if that information were available?
4. What measurable value could result from that change?

Each interview should then address implementation and political feasibility:

5. Who would benefit from the proposed architecture, and who might believe they would be harmed?
6. What is the strongest substantive argument against it?
7. How might the proposal be characterized—or mischaracterized—in public debate?
8. Which concerns could be addressed through architecture, legislation, oversight, or patient control?
9. Which concerns would remain even after reasonable safeguards were adopted?

10. What would skeptical patients, providers, states, and elected officials need to see before trusting the proposal?

Political objections should not automatically be dismissed as misinformation. A fear that the architecture could enable surveillance, healthcare rationing, unauthorized government access, employment discrimination, or a catastrophic breach may reveal a requirement that has not been adequately addressed.

At the same time, the assessment must distinguish a remediable design concern from an emotionally powerful assertion that could be repeated regardless of the evidence.

Adversarial Review

Before completing its findings, the team should conduct an adversarial review of the concept. Participants should be asked to make the strongest possible case against it from several perspectives, including:

- patient privacy and civil liberties;
- federal versus state authority;
- government surveillance or political misuse;
- insurer use of information to restrict care;
- employer, credit, or marketing use;
- cybersecurity and concentrated breach risk;
- federal cost, procurement, and implementation failure;
- disruption of existing healthcare-information businesses;
- dependence on a national service during an outage.

For every challenge, the team should document the underlying concern, supporting evidence, likelihood, potential consequences, and whether it can be addressed through technical design, legislation, governance, auditing, or enforcement. Concerns that cannot be adequately mitigated should remain visible in the final findings.

This is more valuable than developing reassuring talking points. The objective is to strengthen or reject the concept before political advocacy begins.

Consolidating the Findings

AI could assist in organizing the interview transcripts, identifying recurring issues, connecting proposed changes to measurable value, and exposing areas of agreement or contradiction. The team would remain responsible for interpreting the evidence and approving every conclusion.

The findings should include:

- the principal opportunities and credible ranges of measurable value;
- technical feasibility and performance requirements;
- the relationship to TEFCA and existing healthcare systems;
- privacy, security, legal, and governance requirements;
- stakeholders likely to support or resist the concept and their reasons;
- an adversarial risk-and-response assessment;
- issues that remain unresolved;
- a recommendation on whether the concept merits further consideration.

The 60-day assessment should end there. It should not commit the government to a pilot, procurement, implementation program, or particular architecture. Its purpose is to determine whether the opportunity is sufficiently valuable, feasible, and defensible to justify a subsequent decision.

Closing

Healthcare has no shortage of committees studying broad problems.

This proposal calls for something smaller and more disciplined: six qualified people, 60 days, and a specific question.

Can a secure national longitudinal health information architecture give providers timely access to relevant patient history, reduce administrative friction, and create measurable benefits for patients, providers, and payers?

If the answer is no, the reasons should be documented.

If the answer is yes, the nation can then decide what—if anything—to do next.